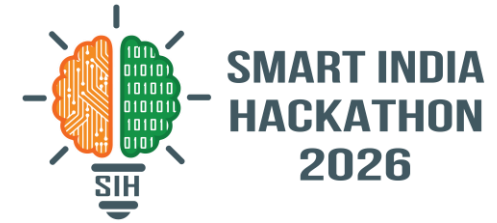


SMART INDIA HACKATHON 2026



SUKOBIN

Carriers as the sensor network

Problem Statement ID – SIH26-MDoNER-01

Problem Statement Title – AI-Enabled Logistics Accessibility Intelligence Platform for the North Eastern Region

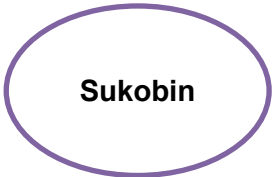
Theme – Transportation & Logistics

PS Category – Software

Team ID – <team id>

Team Name – Sukobin





EVERY VEHICLE ON THE ROAD IS A SENSOR



DETAILED EXPLANATION OF THE PROPOSED SOLUTION

The problem. Landslides, floods and infrastructure gaps cut the NER's roads without warning. Medicines, food, produce and construction material arrive late or not at all, and no integrated platform exists to see it happening.

Nobody drives for us. Anyone already travelling A to B enters their vehicle and route, and is offered only the consignments lying along that road, up to what the vehicle holds. No fleet, no empty return leg.

Those carriers stream GPS while they drive. Median speed against each road's baseline is a live accessibility reading; the weather model turns it into a 72-hour closure forecast.

INNOVATION AND UNIQUENESS OF THE SOLUTION

- **The carrier network IS the sensor network** — one GPS stream both delivers goods and measures the road.
- **Coverage without hardware** — no roadside sensors, no dedicated fleet, nothing to power or maintain.
- **It improves as it is used** — every new carrier adds sensing density at zero marginal cost.

HOW IT ADDRESSES THE PROBLEM — EVERY CLAUSE OF THE PROBLEM STATEMENT

(a) Road & bridge accessibility

42 stretches, 3,567 km, 82 districts, live from GPS

(b) Disruption prediction

24 / 48 / 72 h closure risk on every road

(c) Alternate routes + delay

3 alternatives scored, condition-adjusted ETA

(d) GPS tracking of essentials

medicines, food, produce, construction material

(e) Automated alerts

blocked road, cut-off region, high-risk corridor

(f) Field reporting

photo + GPS + voice, queued when offline

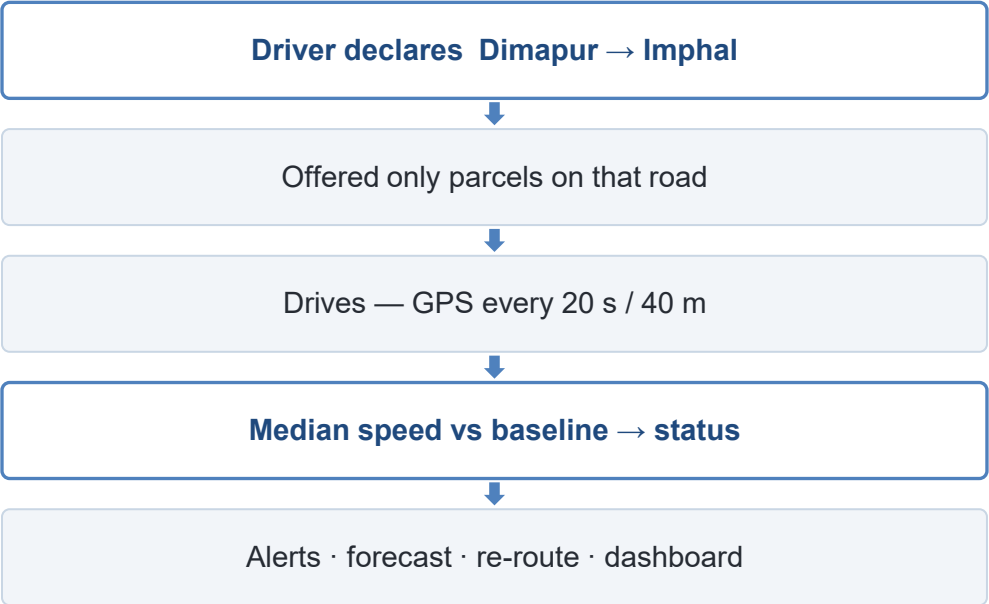
(g) Central dashboard

district status, bottlenecks, emergency routes

(h) Multilingual + offline

10 languages, sync when the signal returns

ONE TRIP, TWO PRODUCTS



TECHNOLOGIES TO BE USED (PROGRAMMING LANGUAGES, FRAMEWORKS, HARDWARE)

SOFTWARE STACK

Backend Node.js · Express 5 · MongoDB with 2dsphere indexes · JWT

Web React · Vite · MapLibre GL for the GIS dashboard

Mobile Kotlin + XML — four Android apps: customer, merchant, driver, officer

Cloud Render · MongoDB Atlas · Cloudinary · Firebase push

AI / ML

Model Logistic regression, benchmarked against gradient-boosted stumps

Features 18 — antecedent rain, burst intensity, slope, terrain, closure history

Trained on 109,116 road-days of observed weather · 42 stretches x 877 days

Scored AUC 0.883 · Brier 0.092 · held out after 2026-03-01

HARDWARE AND EXTERNAL DATA

Hardware None to deploy — the driver's own phone is the sensor

Weather Open-Meteo archive + forecast, hourly

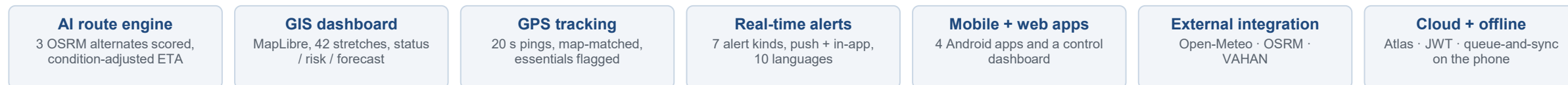
Roads OSRM geometry and route alternates

Transport DB VAHAN registration lookup at driver sign-up

METHODOLOGY AND PROCESS FOR IMPLEMENTATION



EXPECTED SOLUTION — EVERY COMPONENT, AND WHAT IMPLEMENTS IT



Trust rule — a status needs at least 4 samples from at least 2 distinct vehicles, and a driver's report is capped at RESTRICTED until an officer confirms it, so one parked vehicle can never close a highway.

ANALYSIS OF THE FEASIBILITY OF THE IDEA

42 stretches · 3,567 km real OSRM geometry, 12 corridors, 82 districts, 8 states

4 Android apps + dashboard customer, merchant, driver, officer, control room

195 / 195 backend tests passing across 10 suites

10 languages · 8,424 units every screen, zero English fallbacks

100 % coverage status, live vehicle data and forecast on every road

VIABLE BECAUSE SCALING COSTS ALMOST NOTHING

- No hardware to buy, install, power or maintain.
- A new district needs its road geometry seeded — hours, not procurement.
- Sensing density rises with adoption, at no extra cost per reading.
- Runs on managed cloud services already in use.

POTENTIAL CHALLENGES AND RISKS

No public register of past road closures

One parked driver could read as a closure

Hill districts drop off the network

Officers and drivers do not share a language

A quiet road has no vehicles to sense it

STRATEGIES FOR OVERCOMING THESE CHALLENGES

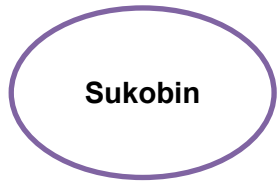
Labels drawn from a rainfall-threshold hazard function; verified field reports override them, and the dashboard states this openly.

A status needs ≥ 4 samples from ≥ 2 distinct vehicles, else the reading is discarded and the road keeps its previous status.

Reports queue on the phone under a client ID and sync later; the server rejects duplicates, so nothing is filed twice.

Ten languages including Khasi, Mizo, Nagamese and Kokborok — spoken aloud as well as written, so literacy is not a barrier.

The weather model forecasts every road regardless of traffic, and the dashboard shows honestly what is sensed and what is not.



IMPACT AND BENEFITS



POTENTIAL IMPACT ON THE TARGET AUDIENCE

District administration A lifeline corridor's closure risk 72 hours out, ranked by how many people lose their only road.	Field officers One screen: weak points ranked with the reason each scored, and a queue of reports to confirm.	Drivers and small operators Income from a journey already being made — no fleet, no contract, no empty return leg.	People in remote blocks Medicines, food, produce and construction material arrive — and an alert in the language spoken at home when they will not.
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BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL, ETC.)

Social	8 states · 82 districts. Alerts and voice reporting in 10 languages, four of which have almost no other software support.
Economic	No dedicated fleet, so moving one more parcel costs close to nothing. Spare capacity in vehicles already on the road becomes income.
Environmental	Fewer dedicated trips — a parcel rides in a vehicle that was making the journey anyway, instead of a second one setting out.
Governance	Every incident carries a photo, a GPS fix, a timestamp and the officer who confirmed it — an evidence trail, not a phone call.

Improved regional connectivity · stronger emergency response · fewer supply-chain disruptions · planning and monitoring on evidence

DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

DATA SOURCES THE PLATFORM RUNS ON

Open-Meteo Historical Weather API

open-meteo.com/en/docs/historical-weather-api — observed hourly rainfall, snowfall and temperature; the model's training set

OSRM — Open Source Routing Machine

project-osrm.org — road geometry and route alternates, 42 stretches

VAHAN / Parivahan registration lookup

parivahan.gov.in — vehicle class and capacity at driver sign-up

MapLibre GL JS + Esri Dark Gray Canvas

maplibre.org — the GIS layer of the control dashboard

PROBLEM AND DOMAIN RESEARCH

MDoNER — problem statement and regional context

mdoner.gov.in — NER connectivity gaps and infrastructure priorities

NHIDCL / MoRTH national highway network

nhidcl.com · morth.nic.in — corridor alignments and NH numbering used to seed the road network

IMD rainfall climatology for the North East

mausam.imd.gov.in — monsoon windows and the rain thresholds behind the hazard function

GSI landslide susceptibility mapping

gsi.gov.in — which stretches are treated as landslide-prone

Honest note on the model — no public register of past NER road closures exists, so historical labels are drawn from a rainfall-threshold hazard function calibrated on IMD and GSI data. Verified field reports override every drawn label, and the dashboard says so on screen.